

Urbanization Prediction from Sentinel-2 Imagery Using Deep Multispectral Regression

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Abstract

We present a modular deep learning pipeline for predicting urbanization intensity from Sentinel-2 multispectral imagery. Leveraging TorchGeo pretrained backbones and custom spectral index engineering, we trained a ResNet50-based regression model on over 600,000 samples with performance surpassing traditional models. We also benchmarked against XGBoost and Vision Transformers. A streamlined training system using Albumentations, early stopping, and fast CSV-based workflows ensured reproducibility and efficiency. Code and data management were organized for flexibility, and all training visualizations and logs were exported automatically. Our experiments show that extended spectral input yields marginal but consistent performance improvements, particularly near urban-rural boundaries.

1 Introduction

Urban centers are hubs of education, healthcare, transportation and work. People thrive in cities as the aggregated number of opportunities arises due to the exchange between individuals and organizations. Over the past 40 years, there has been a steady global shift toward urban living. Around 2009, the world reached a milestone where the number of people living in urban areas equaled those in rural regions [16]. This trend is expected to continue, and by 2050, approximately 70% of the global population is projected to reside in cities. More detailed analysis show that the trend vary across countries but it is particularly strong in Asia, South America and Africa. For instance, some Shared Socioeconomic Pathways [14], global scenarios to support climate change research, indicate that Nigeria, Uganda, India, China, and most parts of Latin America and the Caribbean will undergo a steep increase in urban population to top 90% by 2100. In Mexico, the proportion of urban population reached 81% in 2020 [4], perhaps exacerbated by the effects of climate change in terms of floods, and droughts, all associated with irregularity rainfall cycles. Similar to the global counterpart, as they become more or less attractive in terms of security and environmental conditions, some urban areas become more or less attractive, to the degree that while the general trend is form them to grow, some of them may actually be shrinking [5]. The discrepancies in the corresponding of urbanization make it important to study at a level of resolution suitable for decision making.

In this paper, we are interested in the characterization of urbanization and the degree at which a community can be assigned a degree of urbanization. That is, in the past, the split between what is rural and urban was usually established by population thresholds. For instance, the Degree of Urbanisation, endorsed by the EU and the UN, uses a population grid and density thresholds to identify cities, towns and rural areas [7]. In this classification, a urban centre has a population size of more than 50,000 inhabitants and a density of more than 50,000 inhabitants per square km, while a rural cluster consists of a human settlement with between 500 and 4,999 people with a density larger than 300 inhabitants per square km. Yet, it has been long recognized that urban and rural are not neatly defined. Waldorf [19] introduced the Index of Relative Rurality (IRR), a blend of population size, density, land use, and distance to services to place areas in a rural-urban continuum.

Recently, INEGI released the ITUR dataset, a multidimensional approach integrating demographic and territorial variables to estimate the degree of urbanization at a granular level specified by the Statistical and Geographical System Level 9 (cells of about 460 meters per side) [12]. Along the demographic dimension, the new index integrates population size and density. On the territorial dimension, it considers distance to localities with more than 50,000 inhabitants, the housing deprivation index, land use proportions between productive/vegetation and built-up areas, urban equipment access (proximity to schools and health care centers), and accessibility (distance to paved roads and transportation infrastructure). The exact algorithm is proprietary and not fully published in academic sources. However, the general steps require standardization of variables, dimensional weighting, aggregation at statistical meshgrid cell level [13] and scaling to values between 0 (completely rural) and 100 (completely urban). As it requires data from the general census of population and housing, ITUR is likely to be updated every ten years.

The combination of Machine Learning (ML) and Remote Sensing (RS) has resulted in the possibility to real-time, long term and accurate monitoring the development of phenomena applied to precision agriculture, disaster management, wild life conservation, climate change monitoring and urban change detection. This research explores to what extent ML and RS can be leveraged to temporal resolution at which the ITUR index can be estimated. The contributions of this paper are the following:

- It provides the first assessment to the extent at which ITUR can be estimated using multispectral satellite images at 460 m geostatistical meshgrid level.
- Using a battery of methods ranging from classical to attention based and using the multichannel bands of Sentinel-2 and spectral indices, this research provides a performance base line.
- The above results in a method which increases the temporal resolution at which the ITUR can be obtained from every ten years to basically the next satellite pass.
- The code and instructions are made available for others researchers to verify the results and use them as stepping stones to further advance the performance.

The rest of the article develops as follow. First, we review the related literature at the intersection of urbanization characterization, machine learning and remote sensing. Then,

we describe the approach followed to estimate the ITUR index from ML and RS. This description is followed by a detailed description of the application of the methods. Then, we provide a discussion on the relevance of the results in the context of the related research. Finally, we summarize the findings and advance future lines of inquiry.

2 Related Literature

Awuah *et al.* [1] study how land-cover classification accuracy changes along the urban-rural gradient as a function of the spatial resolution and the gradient in landscape structure.

Area built is directly related to the value in the rural-urban continuous. Benchabana *et al.* [3] conclude that building detection needs to be improved. They propose to use in-depth feature extraction and classification of adaptive superpixel shredding.

The ITUR index aims to express the rural-urban relationship in a continuous because human settlements evolve over time. Badreldin *et al.* [2] assess the temporal dynamics of urbanization and cropland in the Nile Delta between 2007 and 2017.

Garcia-Fontana *et al.* [6] analyze the urban area’s expansion over the years and project potential development. In their analysis, they employ Land Use and Land Cover past scenarios and multilayer perceptrons.

He *et al.* [10] propose a ML + RS approach to estimate population in regular grids of 500 m. They use night-time light information and random forest regression.

Hao *et al.* [8] present the result of studying multitemporal trends in LULC in Lahore, Pakistan. They focus on urban growth, vegetation cover, and barren areas basing their analysis on the employment of multispectral indices related to vegetation, moisture, built-up areas, and water indices.

3 Repository and Pipeline Overview

Our implementation is hosted at <https://github.com/daniel0tgc/ITUR-Prediction-ML>. It supports:

- Flexible channel selection via the `custom_channels` variable
- TorchGeo pretrained backbones (ResNet, ViT)
- Albumentations-based augmentation
- Early stopping and automatic metric logging
- Output management via a structured `outputs/` folder

The training logic is modular, supporting both the legacy slow-loading mode and fast CSV-parsed mode. The project structure emphasizes reproducibility and experimentation.

4 Methodology

4.1 Data and Preprocessing

We use Sentinel-2 Level-2A imagery, aligning and resampling all 13 native bands to 10m resolution.

Our pipeline also computes and normalizes several indices, including NDVI, NDBI, SAVI, MSAVI, NDWI, NMDI, and NBR using the `spyndex` library [15]. Custom feature extraction is modularized for fast experimentation. These spectral bands were chosen due to their rich use cases. NDVI provides vegetation health and density information while NDBI has been used to map out built-up areas. Each of the bands provides more information relevant for our use case by calculating new values using the base 13 bands.

To accelerate training and reduce I/O overhead, we preprocess the data into CSVs with matched image paths and ITUR values. These include `train_matched_data.csv`, `val_matched_data.csv`, and `test_matched_data.csv`, each referencing imagery from a local folder of 96×96 .tif patches.

Prior to pre processing the data, we matched the `celda` code from each image to the corresponding ITUR value in the ITUR dataset. This results in a composed dataset containing pairs of image with thier corresponding true ITUR value.

4.2 Models and Training Configuration

We evaluate three different modeling approaches for predicting urbanization intensity/ITUR values from the processed Sentinel-2 data: a CNN-based deep neural network, a transformer-based deep neural network, and a gradient-boosted decision tree model. The CNN model is based on the **ResNet-50** architecture [9], which consists of 50 layers with residual skip-connections that enable training of very deep networks. ResNet-50 has been widely used in computer vision and has proven effective for remote sensing image classification as well [11]. In our implementation, we initialize the ResNet-50 with weights pre-trained on a large geospatial dataset through the `TorchGeo` library [17]. `TorchGeo` provides models pre-trained on multispectral satellite imagery (including all Sentinel-2 bands), which allows us to leverage transfer learning on our task [18]. Specifically, the first convolutional layer of ResNet-50 is adapted to accept the Sentinel-2 spectral channels (rather than just 3 RGB channels) and the model was initially trained on a broad land-cover classification corpus. Using this pre-trained model as a starting point, we fine-tune all layers on our urban data. The learning rate is set relatively low during fine-tuning to preserve useful feature representations learned from the source data while still adapting to our specific prediction target.

Vision Transformer `vit_small_patch16_224`. For our transformer baseline we adopt the `vit_small_patch16_224` variant provided by the `timm` and `TorchGeo` libraries. Rather than training from scratch, we load the ImageNet-21k-pre-trained weights released with `vit_small_patch16_224`. Traditional CNNs have a built-in locality bias: features are aggregated hierarchically, so long-range context is only available in deep layers. Urban patterns, however, often span hundreds of metres—e.g. zoning transitions and settlement morphology. The self-attention mechanism in ViT allows every patch token to attend to

every other patch in a single layer, capturing global spatial dependencies that can improve urban–rural boundary delineation and detect dispersed settlements that CNNs may overlook.

We resize each 64×64 Sentinel-2 patch to 224×224 and fine-tune the network for 40 epochs and initial learning rate 3×10^{-5} . As with the ResNet, we adopt a 50/20/30 train/validation/test split. Early stopping (patience of 5) is triggered on validation R^2 . However, due to memory constraints, we limited our test to 100,000 images rather than the exhaustive dataset as used in the ResNet50 tests.

XGBoost on the other hand, operates on handcrafted features from each image patch. Training failed on full data due to out-of-memory issues (188,601 samples, 13,248 features), so we used 100,000 samples and limited `n_jobs=1`. We utilized `XGBRegressor` from `xgboost` library.

5 Training Details

Our main training script, `ML_Updated.py`, supports both legacy and fast workflows. Major settings:

- Batch size: 16 (used for all ResNet tests. Used 8 for XGBoost and ViT)
- Epochs: 40 (early stopping with patience of 5)
- Learning rate: 0.0001
- Augmentation: Horizontal/vertical flips, random brightness
- Output: model weights, loss/ R^2 plots, and metrics

Models were trained on a GPU-enabled machine. ViT and XGBoost required subsampling to 100,000 images due to memory constraints. Training logs and images were saved to `outputs/`.

6 Results

Model	Spectral Bands	R^2	MAE / MSE	Sample Size
ResNet50 (baseline)	13	0.6838	0.121 / 0.0218	Full dataset
ResNet50 + indices	19	0.689	0.118 / 0.0212	Full dataset
ResNet152	13	0.681	0.1225 / 0.022	Full dataset
XGBoost (100k)	All features	0.57	N/A	100k
ViT (100k RGB)	19	0.6202	0.124 / 0.0269	100k

Spectral indices slightly improved urban edge delineation. However, spectral indices showed improved initial performance that later began to overfit, roughly around epoch 15. ViT underperformed due to resolution scaling and limited compute. ResNet152 began overfitting, likely due to the immense amount of parameters. The validation loss also began to stop improving early, likely a result from overfitting. The test with no spectral bands utilizing ResNet50 weights showed little sign of overfitting and benefited from early stopping.

7 Training Visualizations

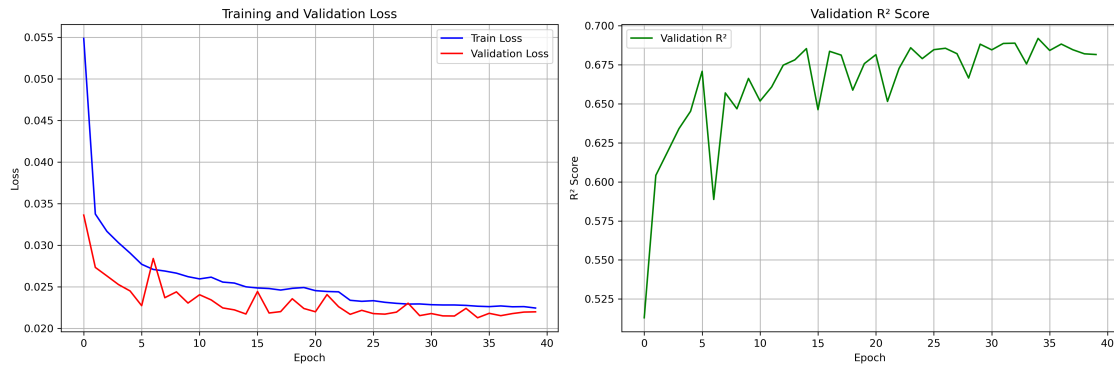


Figure 1: ResNet50 with 6 extra indices (16 channels).

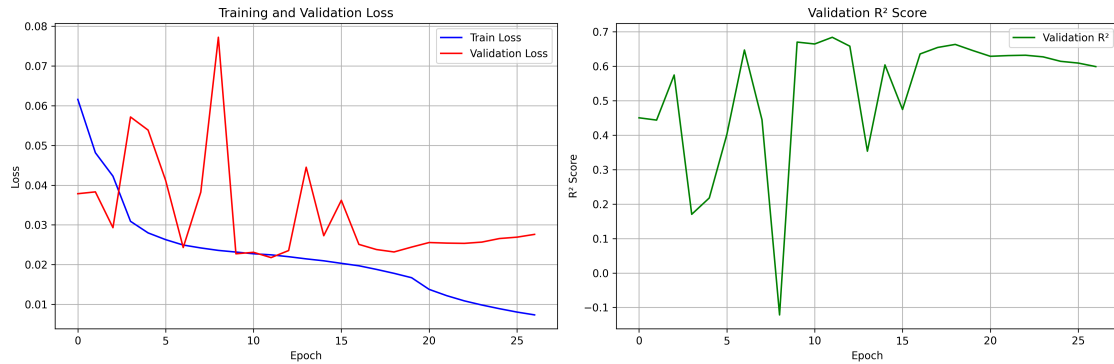


Figure 2: ResNet50 baseline training and validation curves.

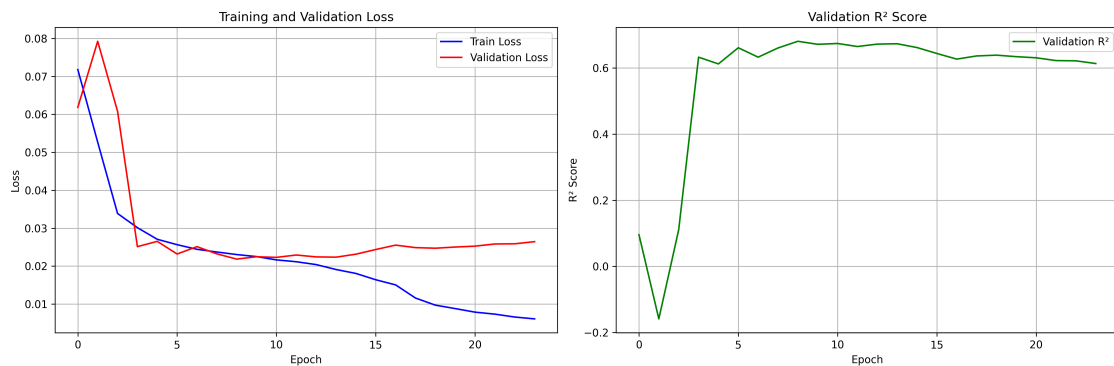


Figure 3: ResNet152 performance with full 600k samples.

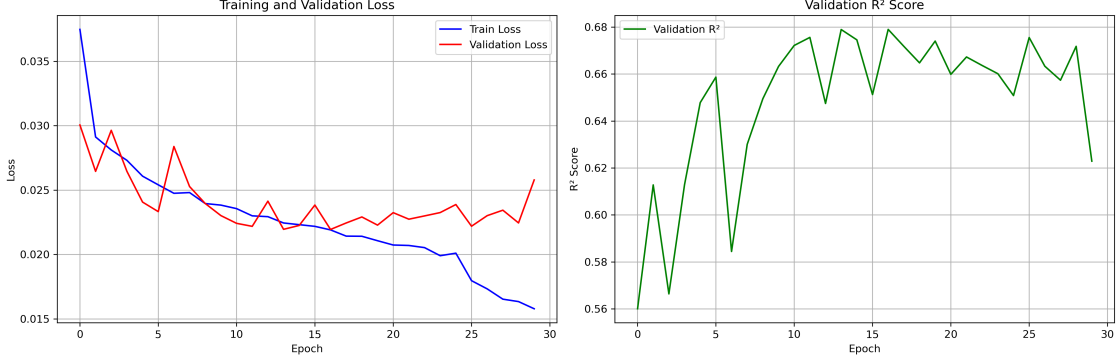


Figure 4: ViT model trained on 100k RGB upscaled patches.

8 Discussion

The results of our study reflect the nuanced trade-offs between model complexity, data richness, and computational constraints in the context of urbanization prediction. ResNet-based architectures performed consistently well, benefiting from both pretraining and the integration of additional spectral indices. While the performance gain from adding indices such as NDVI, NDBI, and MSAVI was modest, it proved consistent—suggesting that these indices help encode urban land cover patterns not immediately captured by raw bands alone.

The ViT-based approach underperformed relative to CNNs, which is attributable to several factors. First, the ViT model required upscaling the 64×64 input patches to 224×224 resolution, which introduced spatial distortion and blurred edge features critical to urban–rural differentiation. Second, limited computational resources restricted training to a subset of 100,000 images, likely constraining the model’s ability to generalize. Nonetheless, the ViT architecture remains a compelling direction for future work, particularly with access to more computational resources and patch-wise attention optimization.

XGBoost, while effective on a handcrafted feature space, struggled at scale. Memory limitations forced us to downsample the dataset and reduce parallelism, curtailing its predictive potential. That said, its relative simplicity and fast inference time make it a candidate for lightweight deployment in resource-constrained settings.

The use of precomputed CSVs eliminated the expensive image/label matching step. Error handling, flexible channel selection, and early stopping enhanced model reproducibility. The addition of spectral indices provided marginal but valuable gains in spatial granularity, particularly at urban boundaries.

We also observed that training on the full dataset (600,000+ samples) resulted in stronger overall performance but introduced greater noise and variance in validation loss. This suggests a need for stronger regularization or adaptive learning schedules to prevent overfitting to localized spectral artifacts. Additionally, future experiments could incorporate uncertainty estimation techniques or ensemble models to improve robustness.

Overall, the combination of a flexible, modular pipeline with pretrained models and spectral augmentation proved a reliable strategy for predicting urbanization. The experiment’s structure also highlighted the importance of engineering considerations—such as patch resolution, batch size, and logging—on final outcomes. These insights will be crucial as we scale

the approach to other geographies or integrate temporal change detection into the model.

Although our best model utilizing ResNet50 with spectral indices achieved an R^2 value of 0.689, further tests utilizing different methodologies could raise this value further. Perhaps other pre-trained weights can be used that may provide an edge on urbanization prediction.

9 Conclusion

This study demonstrates the feasibility and effectiveness of using deep learning, particularly convolutional neural networks, to predict urbanization levels from multispectral Sentinel-2 imagery. By leveraging TorchGeo’s pretrained models and enriching the input space with domain-specific spectral indices, we observed measurable improvements in prediction performance—especially at urban–rural boundaries. Although the gains from spectral augmentation were modest, they consistently enhanced the model’s ability to generalize.

While XGBoost provided a strong baseline with limited data, and Vision Transformers introduced a promising direction, CNNs such as ResNet50 and ResNet152 remained the most reliable and scalable for this task. The results also highlight the critical role of data volume, preprocessing efficiency, and compute constraints in determining final model quality. Our modular pipeline ensures flexibility and reproducibility for future experimentation, and the visual training artifacts provide transparency into model dynamics.

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